# Chi Zhang

Prospective Ph.D. Applicant for Fall 2027
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## Research Profile

My current research focuses on multi-robot systems, target tracking under noisy measurements, and information-theoretic analysis of localization systems. Motivated by a broader interest in system autonomy and the applicability of control-theoretic ideas, I am currently exploring the intersection of AI and control, particularly the integration of AI-agent workflows with control frameworks to improve the provable reliability and trustworthiness of autonomous systems.

## Education

Southern University of Science and Technology — Shenzhen, China
*M.S. in Control Science and Engineering* — *Sep. 2024 – Jun. 2027 (expected)*
Academic Advisor: Prof. Liangming Chen (*IEEE Senior Member*, Associate Editor of *IEEE T-RO and IEEE T-SMC: Systems*)
Thesis Topic: Target tracking with noisy angle-based measurements.

North China Electric Power University — Beijing, China
*B.E. in Automation* — *GPA: 3.92* ***(Top 1%)*** — *Sep. 2020 – Jun. 2024*
Graduation Honor: **Beijing Outstanding Graduate Award**, awarded by the Beijing Municipal Education Commission.

## Publications

**C. Zhang**, W. Ren, L. Chen, and G. Duan, "Angle-Based Target Tracking Using Kalman Filter and Fully Actuated Control," in ***Proceedings of the 64th IEEE Conference on Decision and Control (CDC)*****, 2025.**

**C. Zhang**, W. Ren, L. Chen and G. Duan, "Angle-Based Cooperative Localization and Fully-Actuated Control for Target Tracking," **under review** in ***IEEE Transactions on Automatic Control*****.** (Full paper)

**C. Zhang** and L. Chen, "Optimal geometric analysis of angle-only target localization," **under revision** following review at ***Automatica*****.** (Regular paper)

**C. Zhang,** L. Chen, M. Fu and L. Xie, "Multi-robot Relative Localization Using Interior Angle Measurements" in ***Proceedings of the 19th International Conference on Control, Automation, Robotics and Vision (ICARCV)*****, 2026.**

**C. Zhang**, MZ. Almuzakki, Y. Wang, and L. Chen, "Optimal Target Localization Using Direction-Only Measurements" submitted to ***IEEE Transactions on Instrumentation and Measurement.***

L. Chen, H. Garcia de Marina, L. Xie, and **C. Zhang** "Multirobot Relative Localization and Formation Control Using Interior Angle Measurements", **published** in ***IEEE Transactions on Control Systems and Technology***.

## Project Experience

**Agentic Workflow for Formalizing Control Theory** — *2026 - Present*

● Developing an agent-based workflow for formalizing results from *Nonlinear Systems* in Lean, organizing definitions, lemmas, and theorems as a dependency graph while preserving the textbook's chapter structure for human navigation.

● Exploring long-horizon agent workflows involving task decomposition, theorem proving, verification, iterative correction, and human oversight, with the broader goal of understanding how reliable AI agents can operate on complex technical tasks.

**State Estimation and Feedback Control under Correlated Uncertainty** — *2024 - 2026*

● Led the development of an angle-based target-tracking framework for two pursuers, from geometric modeling and Kalman estimation to fully actuated control and theoretical analysis.

● Derived a linear time-varying measurement model from interior-angle and self-displacement measurements, identified the process–measurement noise correlation induced by the formulation, and incorporated it into a minimum-variance estimator.

● Established exponential ultimate boundedness in mean square and Lyapunov stability for the estimation error, and designed a fully actuated tracking controller with Lyapunov-based stability guarantees.

**Fisher Information-based Sensing Design and Stochastic Optimization** — *2025 - 2026*

● Identified sensing geometry, beyond estimator design alone, as a fundamental factor in angle-only localization accuracy; formulated the placement problem through Fisher information and derived an exact node-wise decomposition with closed-form optimality characterizations.

● Developed a constrained optimization method based on Langevin annealing and an interior-point barrier, combining analytical optimality characterization with stochastic optimization.

**Reinforcement Learning for Lunar Lander Control** — *Course Project, 2025*

● Implemented and compared five reinforcement-learning algorithms (REINFORCE, Actor-Critic, A2C, Natural Actor-Critic, and Dyna-Q) covering policy-gradient, actor–critic, natural-gradient, and model-based methods, with implementations of policy/value networks, TD and Monte Carlo updates, advantage estimation, natural-gradient computation, and model-based planning.

**Transformer-Based Power Load Forecasting** — *Course Project, 2025*

● Implemented and benchmarked Transformer, LSTM, CNN models for power-load forecasting, building an end-to-end pipeline covering data preprocessing, sequence modeling, hyperparameter tuning, evaluation, and attention-weight visualization and analysis.

## Academic Service & Teaching

**IEEE Control Systems Society (CSS) NextCom Leadership Team** — *2026 - Present*

I serve on the Outreach and Broadening Participation Committee for NextCom, supporting public communications, coordinating external contacts, and contributing to event planning and organization.

**Reviewing Service**

Assisted with manuscript reviews for IEEE T-RO, T-AES, T-SMCS, IROS, CDC, ACC, and ASCC.

**Teaching Assistant**

**Linear Algebra** — *Feb.2024 – Jul.2025*

● Designed homework and exam problems, held office hours, and led tutorial sessions in English.